BRAC UNIVERSITY

Inspiring Excellence

CSE461

Introduction to Robotics Lab

Lab No. : 03
Group : 06
Section : 04
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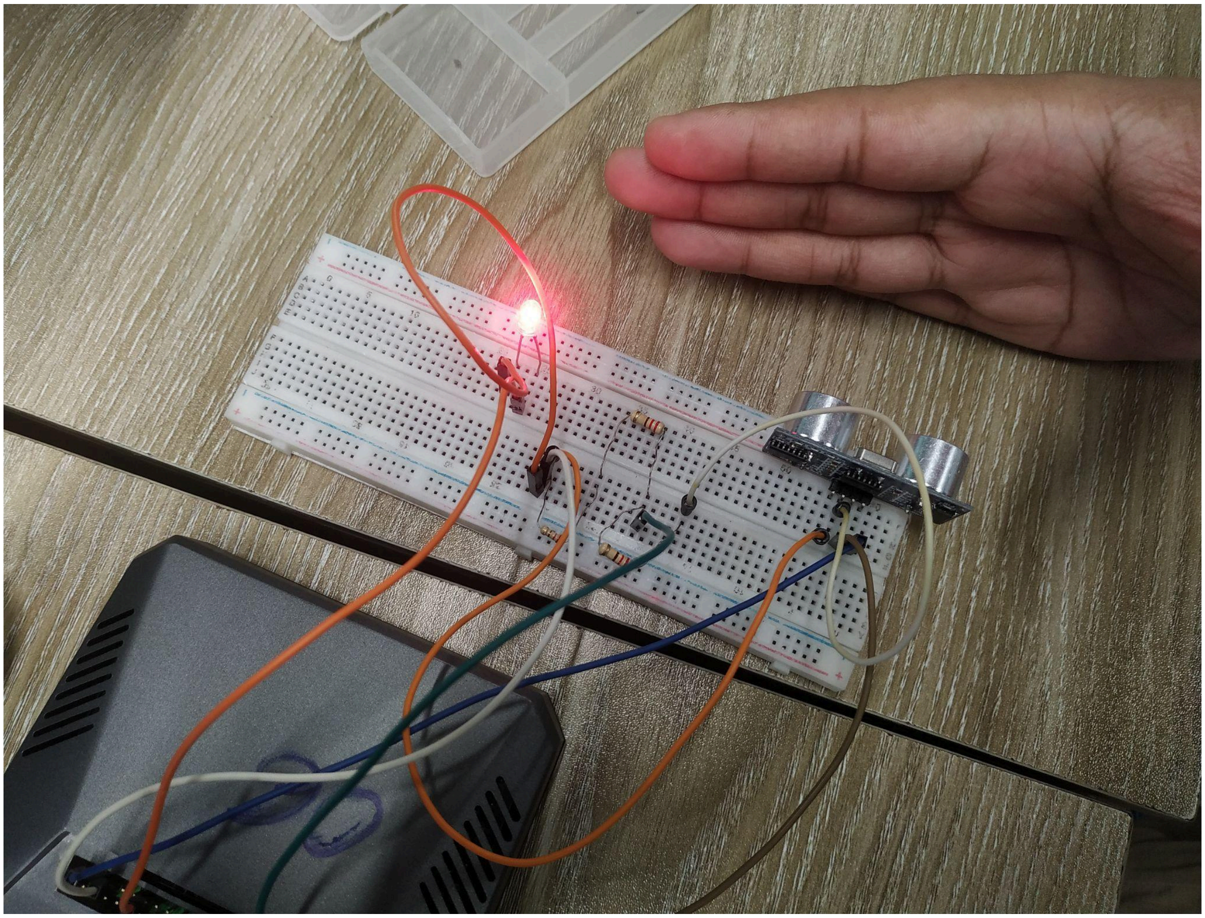
1. **Objectives:** The objective of this lab is to understand the basic functionality of the Raspberry Pi microcontroller and learn how to connect input and output devices using its pins. Specifically, the lab seeks to measure distance with an ultrasonic sensor (HC-SR04) and control the LED depending on the distance measured. Through this experiment, students will have a hands-on experience in circuit-making, interfacing sensors, Python programming on the Raspberry Pi, and implementing simple real-time control logic.

2. **Equipment:**

    - Raspberry PI
    - LED (Light Emitting Diode)
    - Ultrasonic Sensor (HC-SR04)
    - Resistor (220Ω for LED)
    - Breadboard
    - Jumper wires
    - Monitor, keyboard, and mouse (for Raspberry Pi setup)

3. **Experimental Setup:**

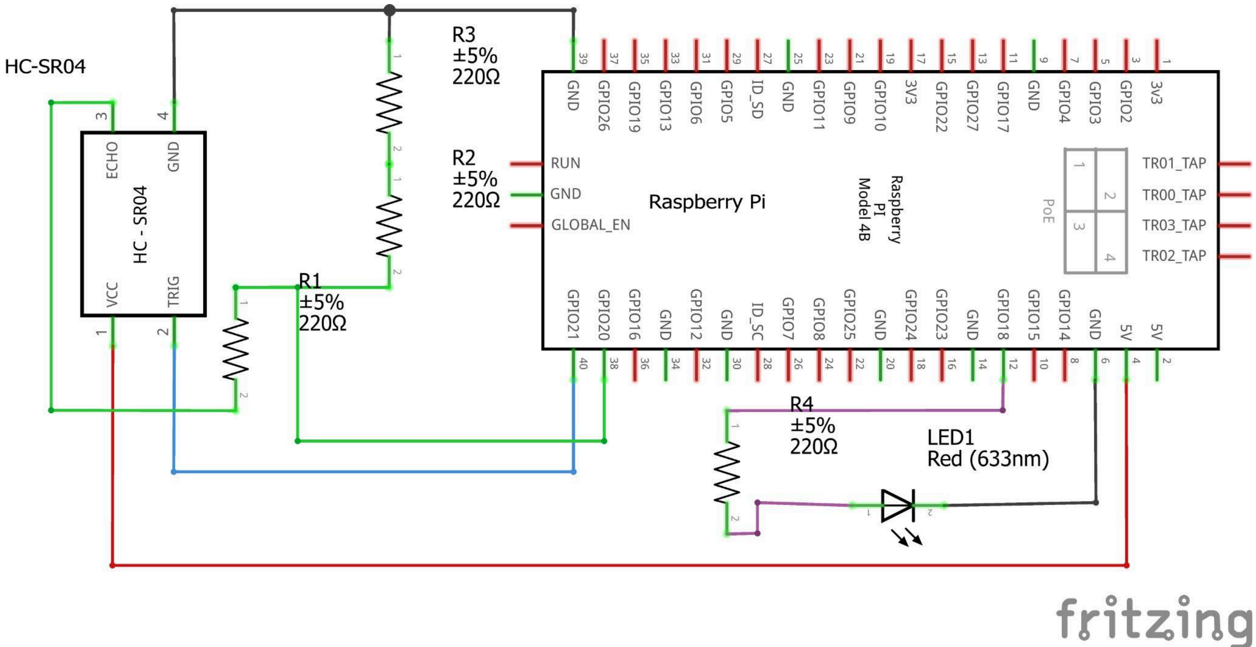


The ultrasonic sensor (HC-SR04) is connected to the Raspberry Pi board using the pin setup, with the TRIG pin connected to pin 21. The ECHO pin is connected to pin 20 using a voltage divider circuit, which is constructed by using 220 Ohm resistors so that the signal does not exceed 3.3V. The VCC pin of the sensor is connected with the 5V power pin, while the GND pin is connected with the ground of the Raspberry Pi.

An LED is connected in series with a 220 ohm resistor to control the current and is connected to pin 18 of the GPIO. The LED will be used as an indicator that is turned ON if the measured distance is less than 5cm.

The control of the circuit is done using a Python program running on the Raspberry Pi. The program is continuously used for sending trigger signals to the ultrasonic sensor to measure the distance by calculating the time delay between the sent ultrasonic waves and the received ultrasonic waves. If the calculated distance is less than 5 cm, then the program turns ON the LED; otherwise, it turns OFF the LED.

Circuit Diagram:



4. Code:

```
import RPi.GPIO as GPIO
from gpiozero import LED, Button
import time

GPIO.setmode(GPIO.BCM)
TRIG = 21  # GPIO21
ECHO = 20  # GPIO20
led = LED(18)

GPIO.setup(TRIG, GPIO.OUT)
GPIO.setup(ECHO, GPIO.IN)

def distance():
    GPIO.output(TRIG, False)
    time.sleep(0.5)
    GPIO.output(TRIG, True)
    time.sleep(0.00001)
    GPIO.output(TRIG, False)
    pulse_start = time.time()
    while GPIO.input(ECHO) == 0:
        pulse_start = time.time()
    while GPIO.input(ECHO) == 1:
        pulse_end = time.time()
```

```
    pulse_duration = pulse_end - pulse_start
    distance_val = pulse_duration * ((343 * 100) / 2)
    distance_val = round(distance_val, 2)
    return distance_val

while True:
    dis = distance()
    print(dis)
    if dis < 5:
        led.on()
    else:
        led.off()

GPIO.cleanup()
```

5. **Results (Output of the experiment):** The experiment was done successfully. The ultrasonic sensor detected the distance between the ultrasonic sensor and an obstacle in real-time, and the LED acted based on the value of the distance. When the object was placed nearer than 5 cm, the LED switched ON, and when the object was moved to a farther distance urging the LED switched OFF.
   During execution, the Python program was continuously sending and receiving the time values and calculating the distance from the formula. For example, the distance detected by the sensor can be read during the testing: 3.45, 4.87, and 6.12 were displayed. When the distance fell below the 5 cm mark, the LED turned on, and we got the successful result of reading the ultrasonic sensor in the Raspberry Pi and controlling the LED output via the pins of the GPIO.

Measured Distance (cm)	LED Output
3.45	LED ON
4.90	LED ON
6.12	LED OFF

6. **Discussions/Answers:**
   This experiment showed how a Raspberry Pi can be used to measure distance with an ultrasonic sensor and turn an LED on/off based on the measured values through the Raspberry Pi's pins. The results showed that the Raspberry Pi successfully received input signals from the ultrasonic sensor and calculated distance accurately in real time. The LED light turned on when the measured distance was less than 5 cm and off when it was

more than 5 cm. So we can ensure that the input device and output device work properly. We faced some issues that we faced during the experiment. For examplethere were some inaccurate readings during the experiment because of the wrong wiring or placement of the register. Overall, the experiment was effective and enhanced learning of Python-based programming for GPIO control, HC-SR04 ultrasonic sensor for distance measurement.